

You are a senior staff-level full-stack engineer.

I already have a working project called DineMate built using:

- Vanilla HTML
- CSS
- JavaScript
- Firebase Authentication
- Firestore

IMPORTANT:

- Do NOT rebuild the project from scratch.
- Analyze the existing codebase first.
- Preserve all existing functionality.
- Refactor and upgrade the existing application.
- Use production-quality architecture.
- Keep Firebase.
- Keep all current features working.

===== PRIMARY GOAL
=====

Transform DineMate into a startup-quality social dining platform.

The final application should feel like a modern consumer product and not a student project.

===== PROJECT STRUCTURE
=====

Refactor code into:

```
src/ ├── auth.js ├── restaurants.js ├── slots.js ├── chat.js ├── notifications.js ├── firebase.js  
    ├── maps.js ├── profiles.js ├── friends.js ├── analytics.js ├── ui.js ├── utils.js └──  
    styles.css
```

Use ES6 modules.

Remove duplicate code.

Improve maintainability.

===== DARK MODE
=====

Dark mode must be the DEFAULT theme.

Requirements:

- Premium dark theme
- Modern gradients
- Consistent design system
- Save user preference
- Allow light mode toggle

===== UI / UX OVERHAUL
=====

Upgrade the entire visual design.

Requirements:

- Modern SaaS-quality design
- Better typography
- Better spacing
- Glassmorphism effects
- Better cards
- Modern navigation
- Premium buttons
- Better forms
- Better profile layouts
- Better restaurant cards
- Better slot cards
- Better mobile responsiveness

Add:

- Skeleton loaders
- Toast notifications
- Empty states
- Smooth transitions
- Hover animations
- Loading indicators
- Page animations

Use:

- Inter font
- Professional color palette
- Consistent shadows
- Consistent border radius

The UI should feel comparable to modern products released in 2026.

===== REAL LOCATION DISCOVERY
=====

Use:

- OpenStreetMap
- Overpass API
- Nominatim API
- Leaflet.js

Requirements:

Automatically detect user location.

Fetch nearby:

- Restaurants
- Hotels
- Cafes
- Food courts
- Canopies
- Dining places
- Fast food outlets
- Coffee shops

Search radius:

- 1km
- 3km
- 5km
- 10km

Show:

- Name
- Address
- Coordinates
- Distance
- Category

Store favorites in Firestore.

Create:

searchNearbyPlaces() searchByLocation() searchByCategory()

===== MAP INTEGRATION
=====

Implement Leaflet.js maps.

Requirements:

Show:

- User location
- Restaurants
- Cafes
- Hotels
- Dining slots

Add:

- Current location button
- Marker clustering
- Category filtering
- Route directions
- Distance calculations
- Open in Google Maps

Markers should be interactive.

===== DEMO DATA
=====

Initialize realistic demo data automatically.

Generate:

Demo Restaurants

Demo Users

Demo Slots

Demo Chats

Demo Ratings

Demo Notifications

The application should not look empty for new users.

===== DINING SLOT SYSTEM
=====

Upgrade slot system.

Schema:

```
{ hostId, restaurantId, date, time, maxCapacity, currentCapacity, attendees, isFull, status, createdAt }
```

Features:

- Join slot
- Leave slot
- Waitlist
- Capacity control
- Slot expiration
- Auto completion
- Auto close when full

Use Firestore transactions.

```
===== FRIEND SYSTEM  
=====
```

Implement complete social system.

Features:

- Follow user
- Unfollow user
- Send friend request
- Accept request
- Reject request
- Remove friend

Collections:

friends friendRequests

Show mutual friends.

```
===== PUBLIC USER PROFILES  
=====
```

Create public profiles.

Example:

User Name

★ Rating

Meals Hosted

Meals Joined

Favorite Cuisine

Favorite Restaurants

Friends Count

Badges

Achievements

Recent Activity

Profile Picture

Cover Banner

Bio

Users can visit other profiles.

===== RATING SYSTEM
=====

Improve ratings.

Allow:

- User ratings
- Restaurant ratings
- Slot ratings

Display:

Average rating

Review count

Recent reviews

===== CHAT SYSTEM
=====

Upgrade chat.

Features:

- Real-time chat
- Read receipts
- Typing indicators
- Message timestamps
- Emoji support
- Chat notifications

===== NOTIFICATIONS
=====

Integrate Firebase Cloud Messaging.

Notifications:

- New friend request
- Friend accepted
- Slot joined
- New message
- Slot reminder
- Slot full

Support:

- Foreground notifications
- Background notifications
- Notification center

===== ANALYTICS DASHBOARD
=====

Create analytics section.

Show:

Meals Hosted

Meals Joined

People Met

Restaurants Visited

Favorite Cuisine

Monthly Activity

Growth Charts

Attendance Rate

===== GAMIFICATION
=====

Add badges.

Examples:

First Meal

Social Explorer

Food Enthusiast

Top Host

Frequent Diner

Display achievements on profiles.

===== SEARCH
=====

Global search.

Search:

- Users
- Restaurants
- Cafes
- Hotels
- Slots

Add filters.

===== SECURITY
=====

Generate production-ready Firestore rules.

Requirements:

- Users edit only their own data.
- Users edit only their own slots.
- Validate ratings.
- Validate friend requests.
- Prevent unauthorized writes.

Provide complete Firestore rules.

===== PERFORMANCE
=====

Optimize:

- Firestore queries
- Realtime listeners
- Rendering
- Memory usage

Use lazy loading.

Use pagination.

Reduce unnecessary reads.

===== NETLIFY DEPLOYMENT
=====

Provide:

- Folder structure
- Build instructions
- Deployment instructions
- Environment variable setup

===== OUTPUT REQUIREMENTS
=====

1. Analyze existing code first.
2. Modify existing code.
3. Do NOT remove existing features.
4. Generate all required files.
5. Provide complete code.
6. Provide Firestore rules.

7. Provide setup guide.
8. Ensure production-ready quality.
9. Ensure mobile-first responsiveness.
10. Ensure dark mode is default.